Cloud-Agnostic Provisioning For Fun and Sovereignty

Cloud-Native Rejekts EU 2026

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Hello, I'm

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TODO list

Why self-host instead of KaaS?

What stack will we use?

Let's get to work (Demo)

Wins and Benefits

Day 2 Sovereignty

Why?

Why?

- Cost Efficiency
- Insurance against Volatility
- Accommodate for existing Changes
- Vendor Independence
- Operational Sovereignty Requirements
- Also, it's fun and we learn a lot!

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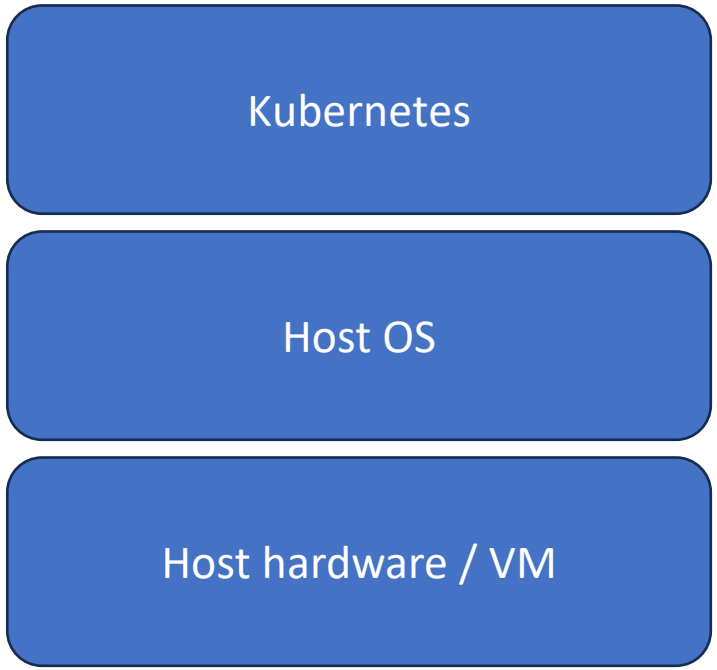
Our Stack

Stack

- FOSS only
- Community Driven
- Flexible and Versatile
- Well Established
- Stable and Secure
- Broad Ecosystem Support
- Easy to operate / maintain

Stack



Host

- Bare Metal
- Private Clouds
  - OpenStack
  - Proxmox
- Public Clouds
  - Biggies – AWS, GCP, Azure, Akamai
  - Mid-size – OVH, Hetzner, Scaleway
  - Regional – Brightbox, STACKIT

Host + Host OS

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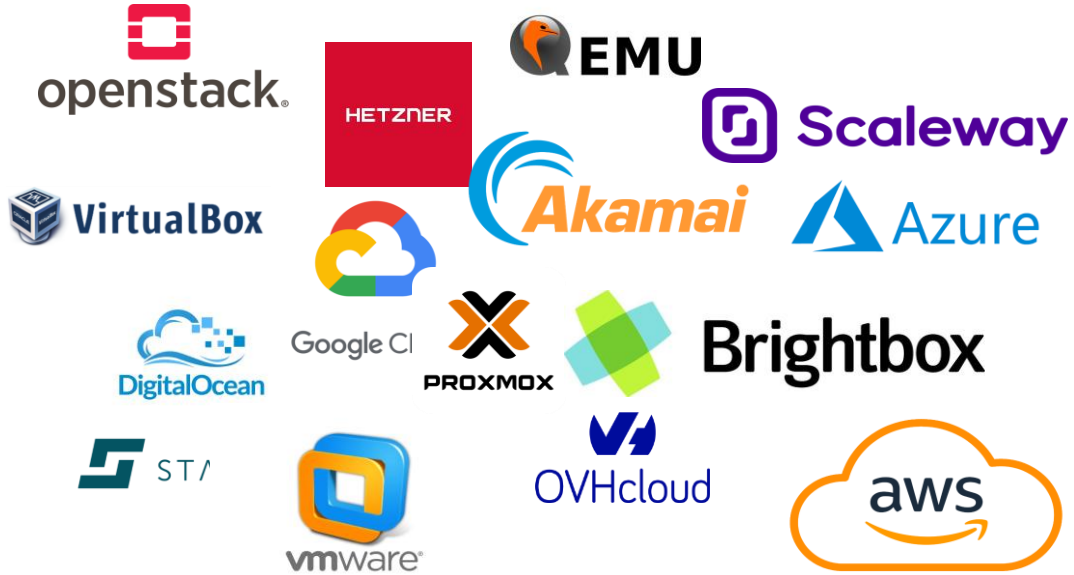
- Broad OS vendor support

- Community Stewarded CNCF project
- Vendor independent configuration
- Well established
- Easy to operate / maintain
- Wide support



Presentation Choices

- Hosts
  - Qemu
  - Hetzner
  - Akamai
- Host OS
  - Flatcar
- Kubernetes
  - Official releases
  - Plain k8s
  - Shipped as immutable, update-able image

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Putting it together

The Demo

- Ummm, how do we demo Sovereignty?

- Let's at least demo cloud-agnostic provisioning
- Provision Kubernetes, "from the trenches"
- Just OS configuration and the Kubernetes image
- Locally, on Hetzner, and on Akamai

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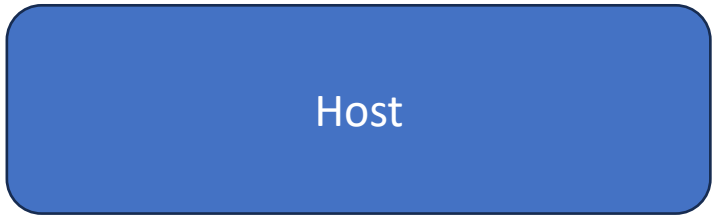

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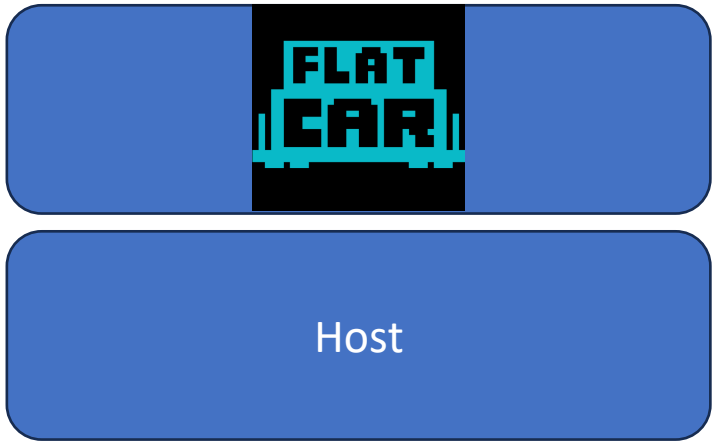

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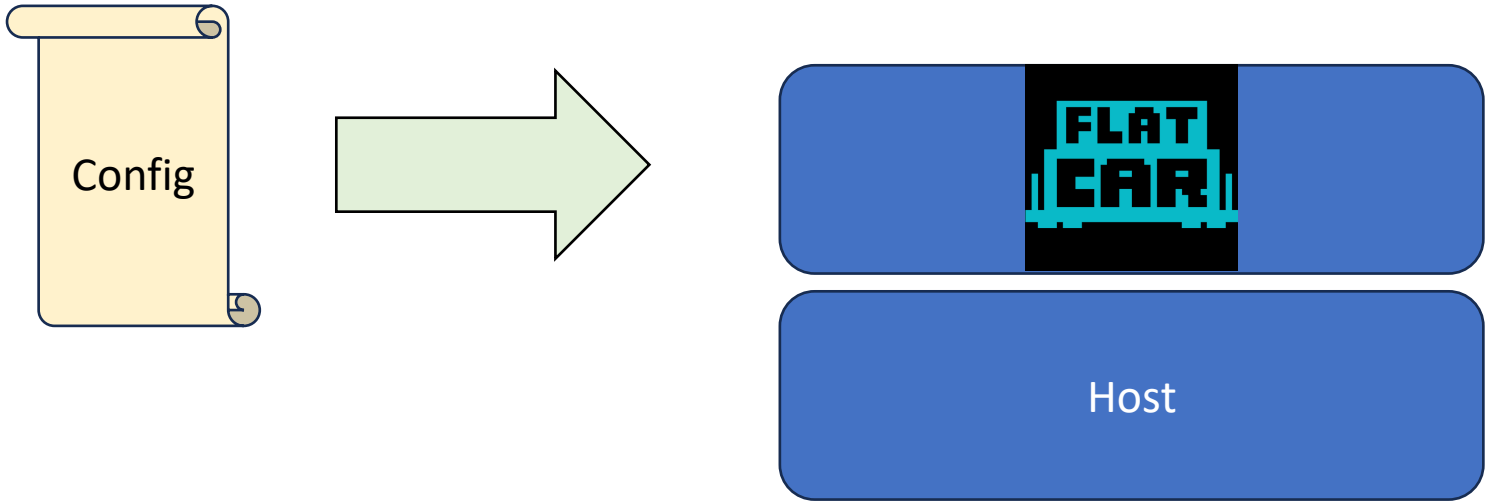

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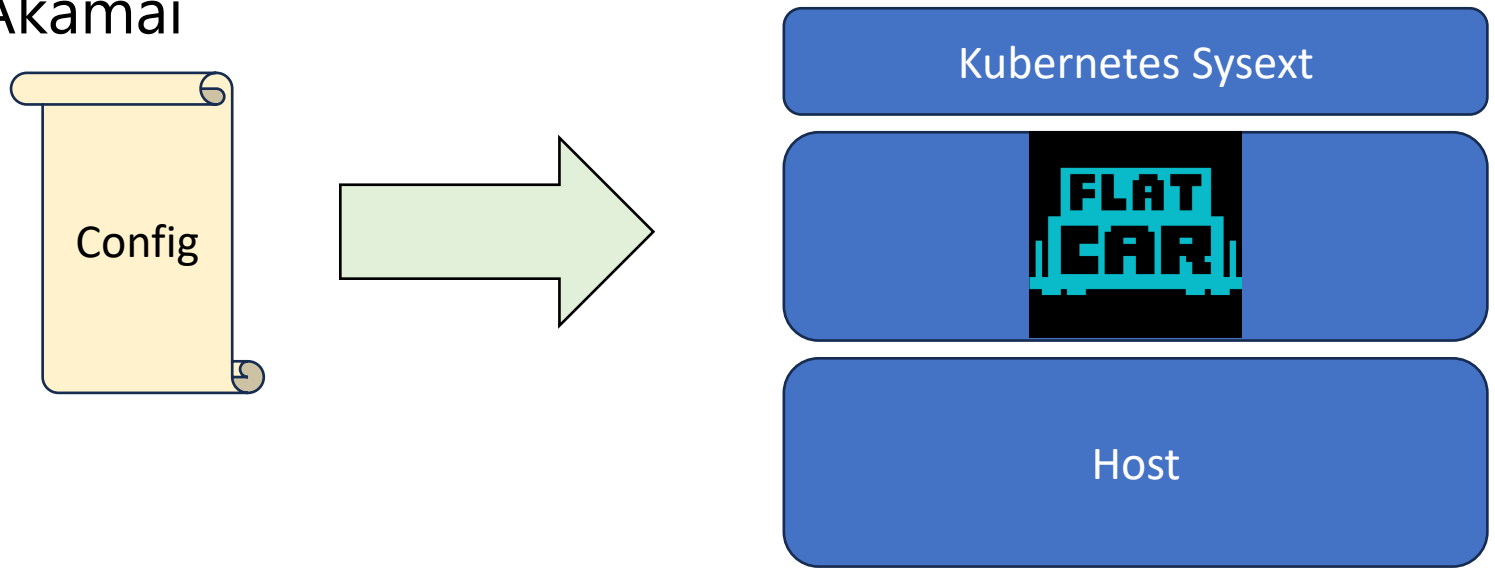

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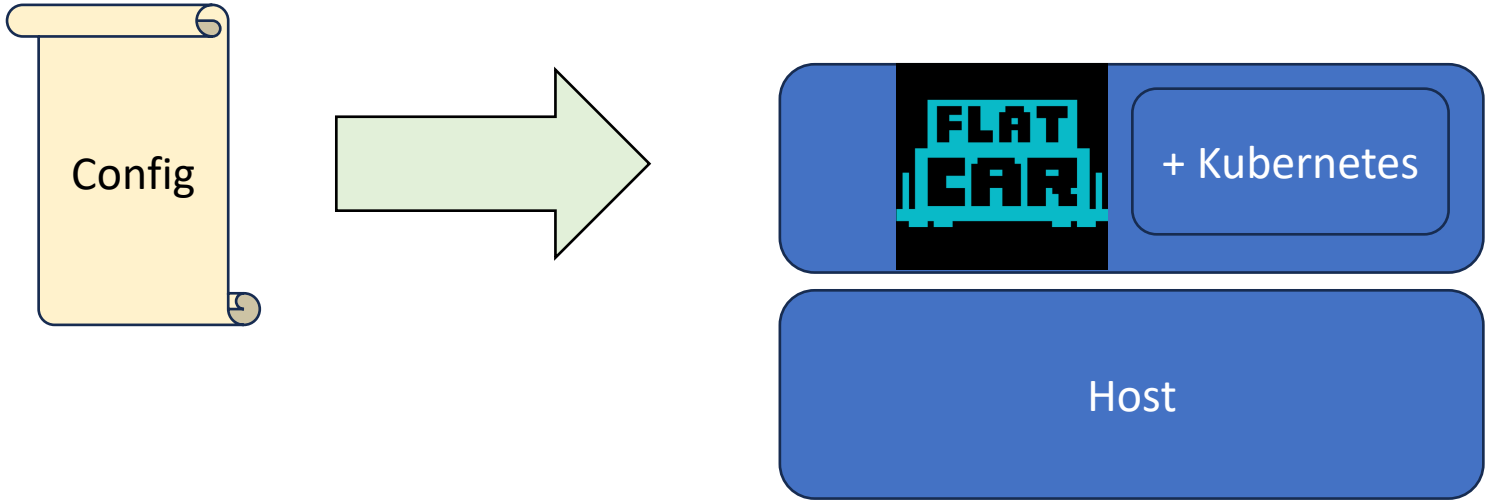

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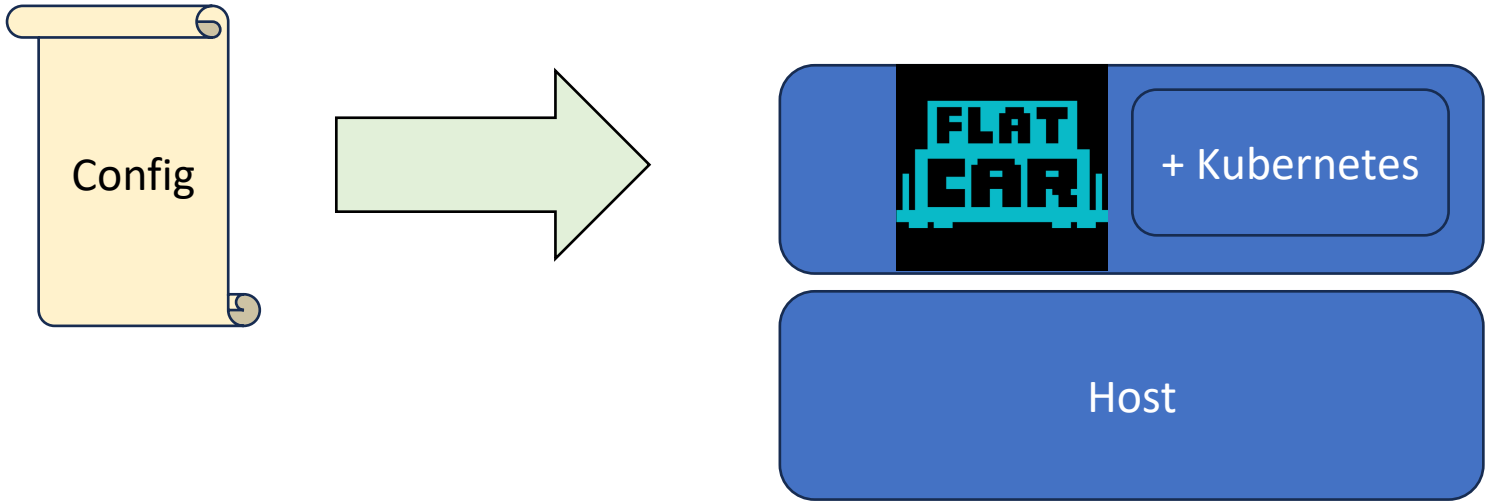

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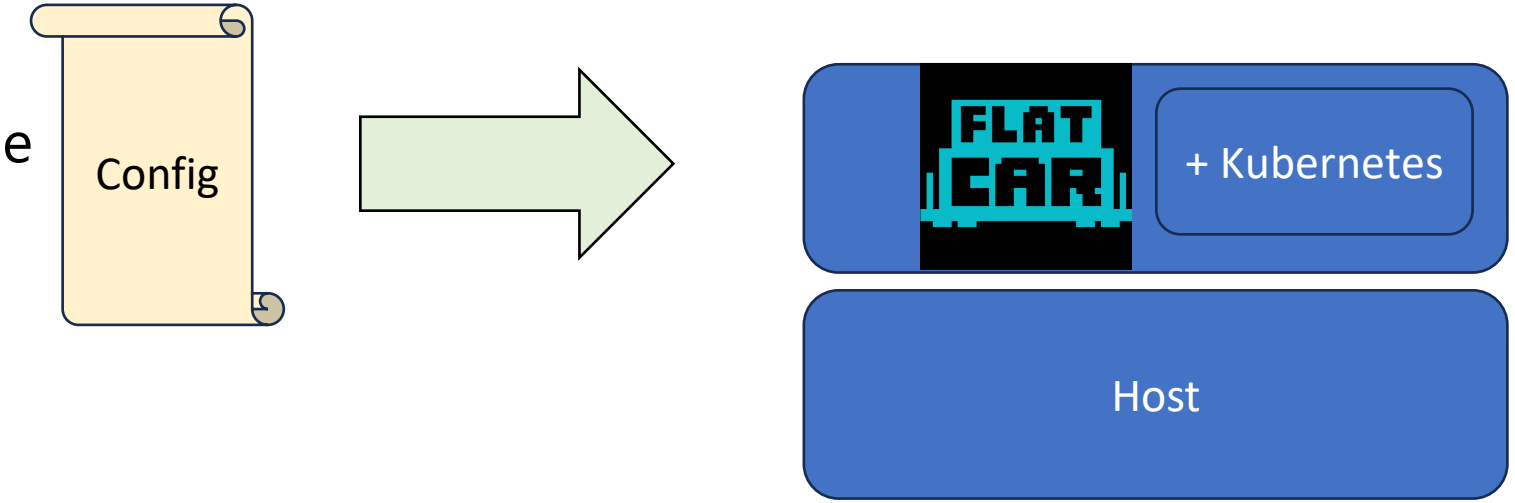
  - Fetch the "join" command
  - Create Workers (auto-join)



QEmu

The Demo

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- Using THE SAME**...ish** config:
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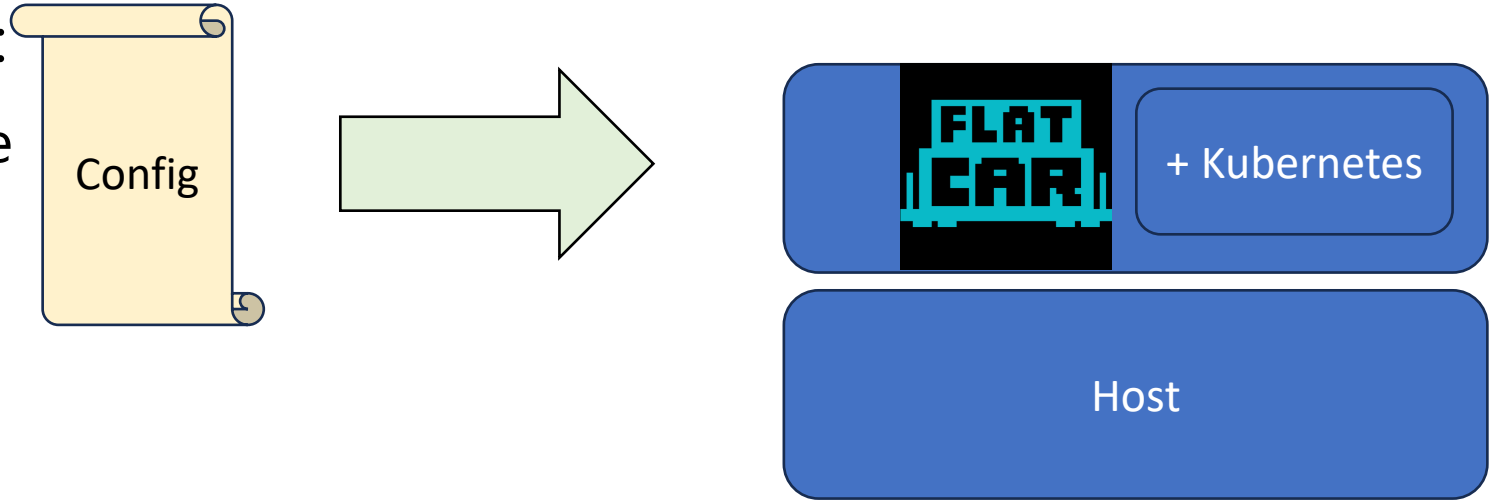

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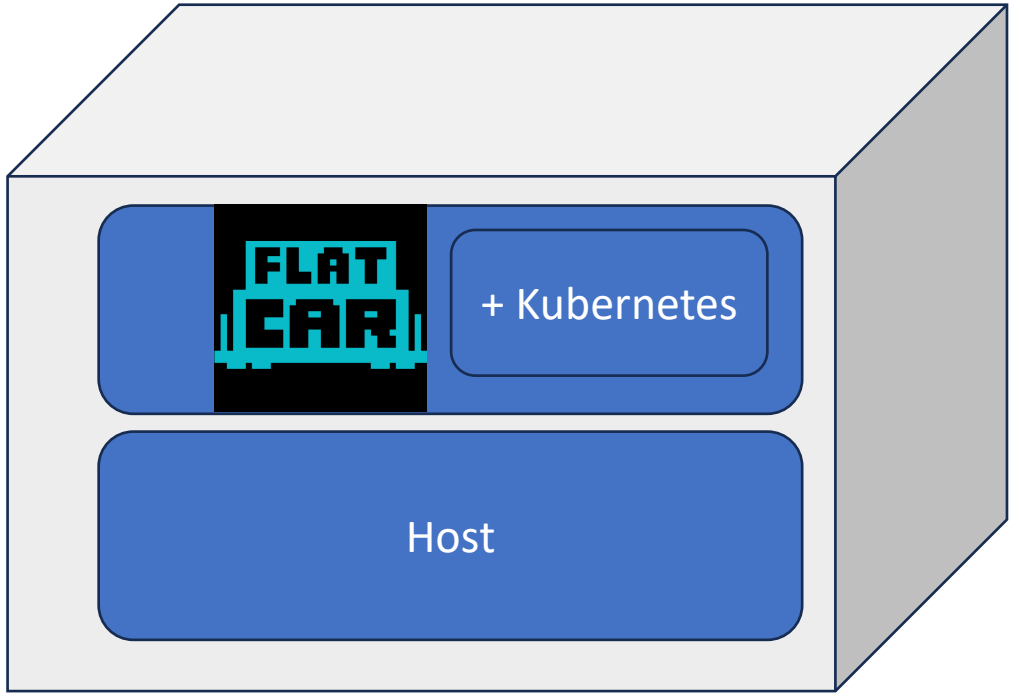

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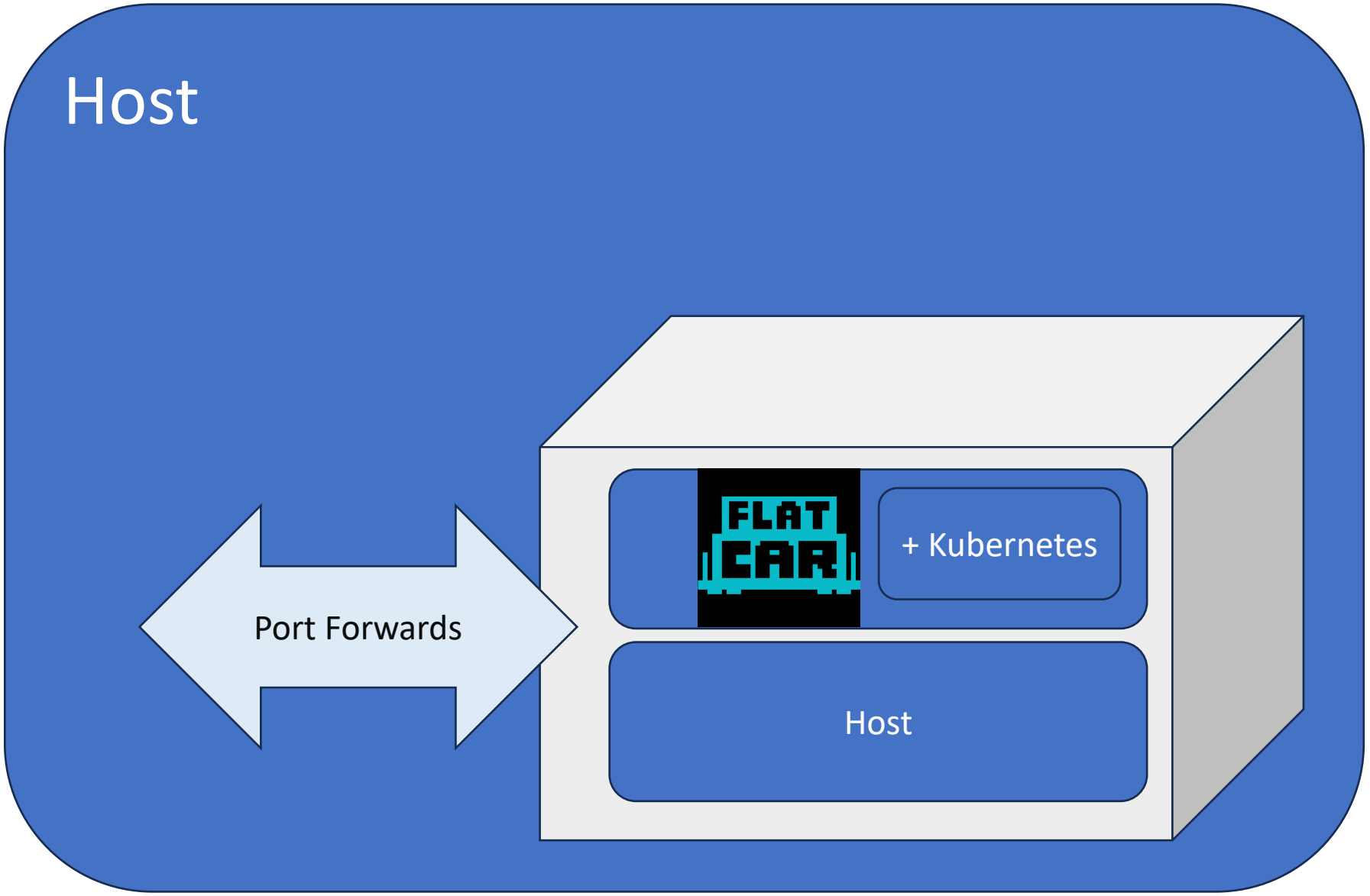

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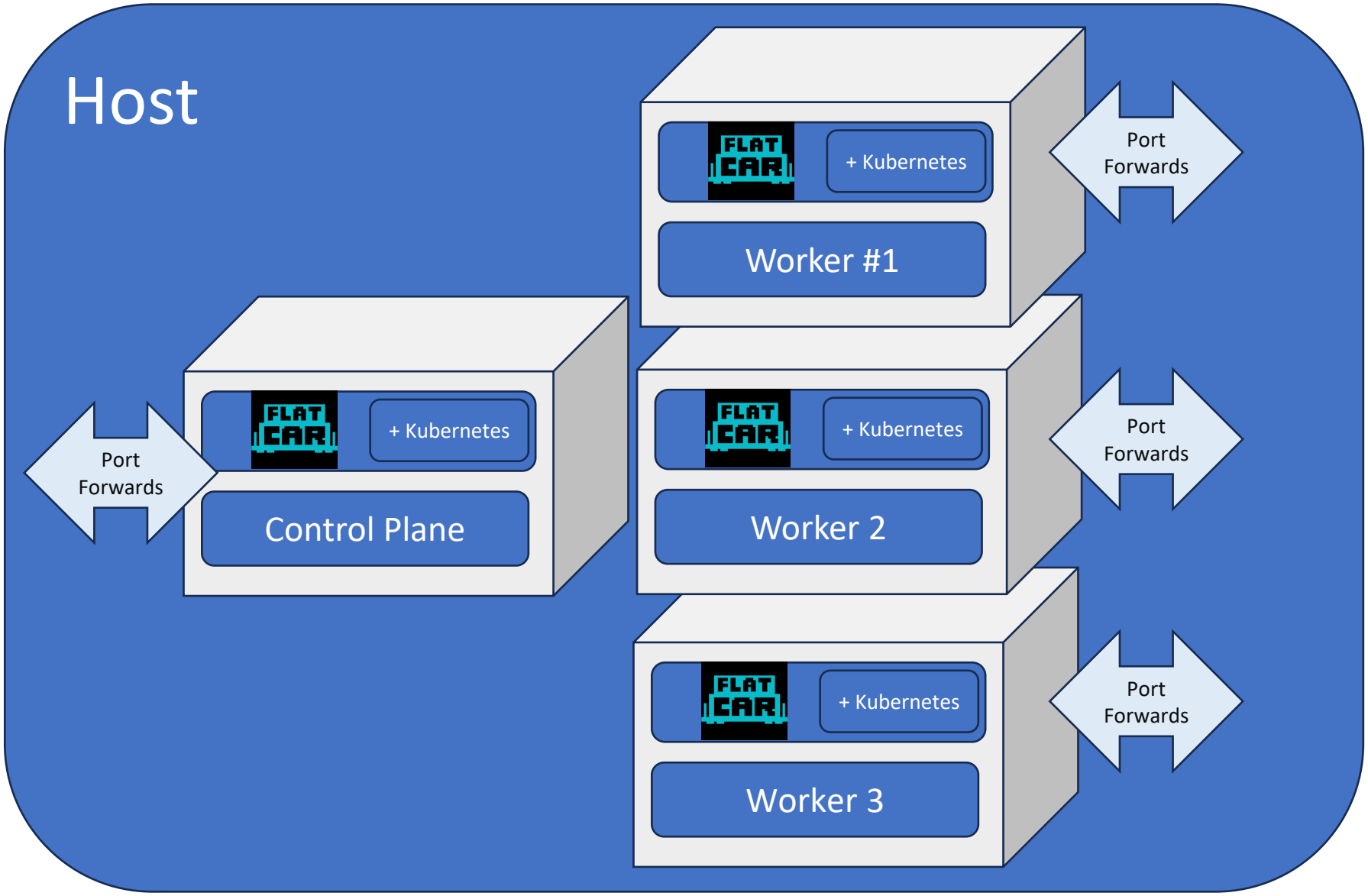
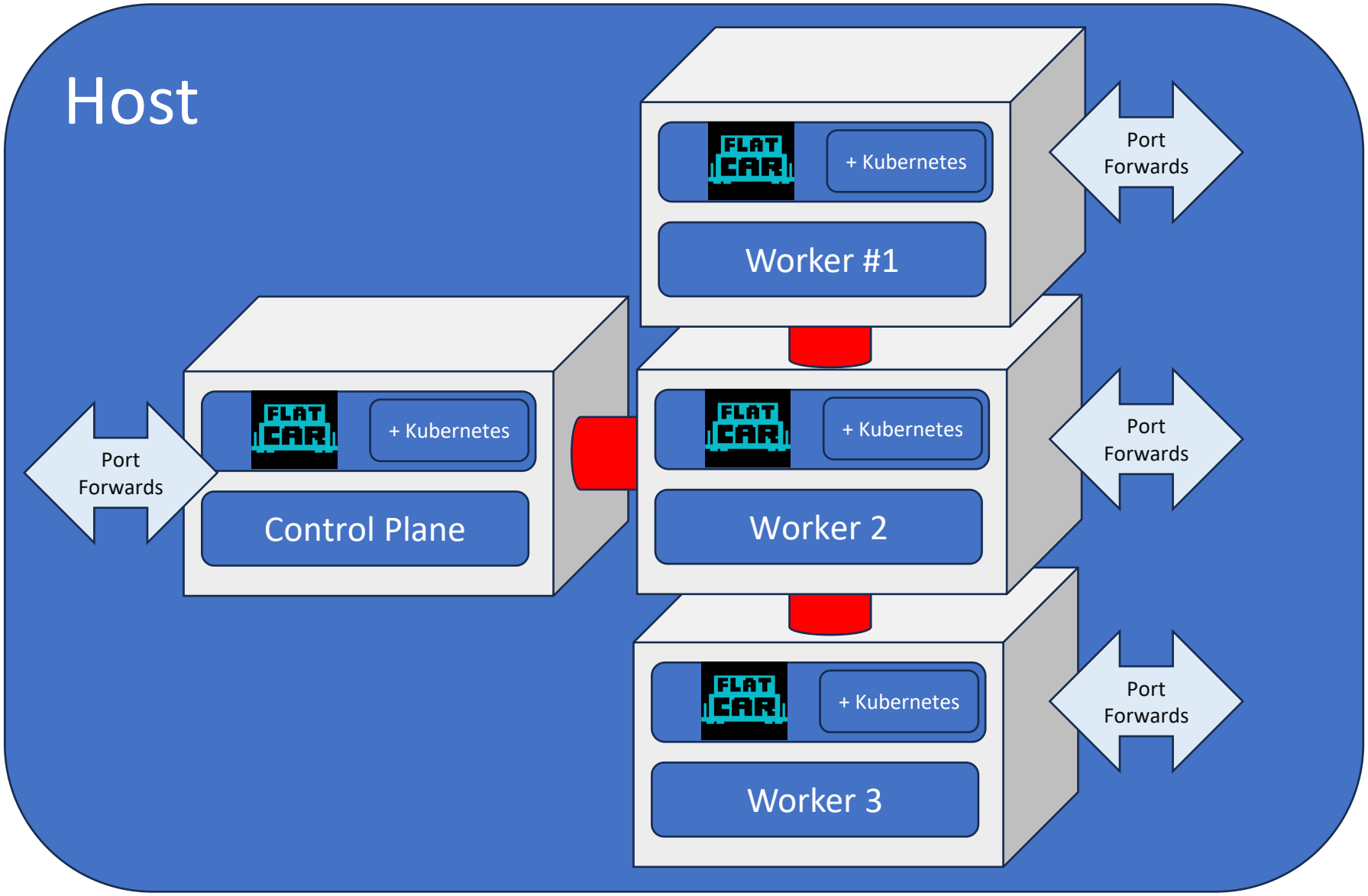


The QEmu Demo uses more complicated networking



Clouds

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Wins and Benefits

Wins

- Many environments supported
  - It's easy to add more!
- Agnostic config == seamless migration
  - Run your nodes where it benefits you most
  - Pricing / size availability
  - Special needs, e.g. GPUs
- Community driven
  - CNCF project, community stewardship
  - Maintainers from most continents (EU centric)
- Well established stack
  - Multi-cloud user base
  - Users build business cases on abstraction
  - OS of choice for hosted Kubernetes offers

Benefits / Opportunities

- Managed In-Place updates
  - Base OS and Kubernetes, independently
- Supports air-gapped environments
  - Easy to host your own infra (updates, sysexts)
- Cryptographically secure
  - Signed OS, signed sysexts
- Easy to extend
  - Upstream likes Patches!
  - We're also very newcomer friendly
  - Deeply customisable via 3rd party sysexts
  - See extensions.flatcar.org

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Day 2

Wait-wait-wait - in-place updates?

- Yupp!
- Separately for Kubernetes and the OS
  - Using the same mechanism
- Demo!

What does *real* Sovereignty entail?

- Can you operate it?
  - Rented service vs. run-your-own vs. mix & match
- Can you keep it stable?
  - Project has strict no-bugs-in-stable policy
  - Use Beta canaries, report issues upstream
- Is it well stewarded?
  - Maintainers: 3 companies + private individuals.
- Can you steward it?
  - Graduation: (even) maintainer diversification
  - Maintainer-driven, you contribute, you steer
- Can you own it?
  - Yes, you can!

Own it

- We pride ourselves to be easily forkable.
  - That's how Flatcar started: as a friendly fork.
- Project is containerised / automated.
  - `git clone https://github.com/flatcar/scripts.git .`
  - `git checkout alpha-4628.0.0`
  - ./run_sdk_container -t ./build_packages
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  - ./run_local_tests.sh
- Thorough test suite
  - Easy to learn, hard to break
- Very few repositories
  - OS, test+release suite, Update Service
- Source caching built-in
  - ./run_sdk_container -t ./update_distfiles

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Thank you